AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I032*

Team Name: *SARAL FORGE*

Domain: *1[Intelligent systems and digital futures]*

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Presentation Outline

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Problem Statement

- The problem that we are solving is to detecting and stopping fake signals , called spoofing of aircrafts or warships intended by attackers to cause intentional damages to infrastructures and civilian livelihood.
- we need this idea right now since,
- After the shift from conventional warfare form battlefields to asymmetric technology driven conflicts, where drones, AI systems changes are used constantly to threaten national infrastructures , civilian livelihoods and the lives of soldiers, It is a matter of concern as India shares 1500 km border line with neighbors and certainly has turmoil in relations with those nations and serious act of national security.
- The Native GNSS anti-spoofing powers Atmanirbhar Bharat by securing India's defense, aviation, and critical infrastructure with tamper-proof navigation security.

Problem Understanding

1. PRIMARY USERS

- Defense and strategic navigation systems
- Aviation and maritime operators
- Transport and fleet operators

2. ECOSYSTEM STAKEHOLDERS

- ISRO–NavIC ecosystem and Indian GNSS technology developers
- GNSS receiver and chip manufacturers
- Safety regulators and integrators of critical-infrastructure

3. NATIONAL BENEFITS

- Safer and more resilient positioning and navigation
- Stronger protection of navigation-dependent infrastructure

Existing Gaps & Challenges

- NavIC's Restricted Service : This service uses secret pseudo noise(PRN) to alter the signals so that only a person with a key to the private PRN can access the signal.
- GAGAN : This solution uses 15 different reference stations to calculate position and a single spoof attack can easily be identified by noticing the deviation from the other 14 reference stations.
  's Regional Boundary Constraint : It only provides coverage across India and up to 1,500 km beyond its borders making it unusable for global navigation outside this region.
- GAGAN's Total Dependency on US GPS : It is merely an augmentation system that relies entirely on underlying GPS signals. Any failure from the GPS systems causes it to fail.
- Indigenous protection: NavIC alone is not spoof-proof—Indian-built secure receivers and anti-spoofing technology are essential.

Our Proposed Solution

- **Core idea :**

  GNN-based anti-spoofing layer for GNSS that detects fake or abnormal satellite signals in real time

  Designed as an indigenous, practical protection solution for defense, aviation, drones, telecom, and critical infrastructure

- **How the solution works at a high level?:**

  It first studies GNSS systems and their spoofing weaknesses, then develops a software-based GNN model to learn unusual signal patterns

  The system is built to flag suspicious navigation data quickly and help reject corrupted signals before they disrupt positioning or timing

- **Main value delivered to users**:

  Faster, real-time protection against spoofing attacks

  Lower-cost and more scalable deployment than heavy hardware-based

Technical Approach

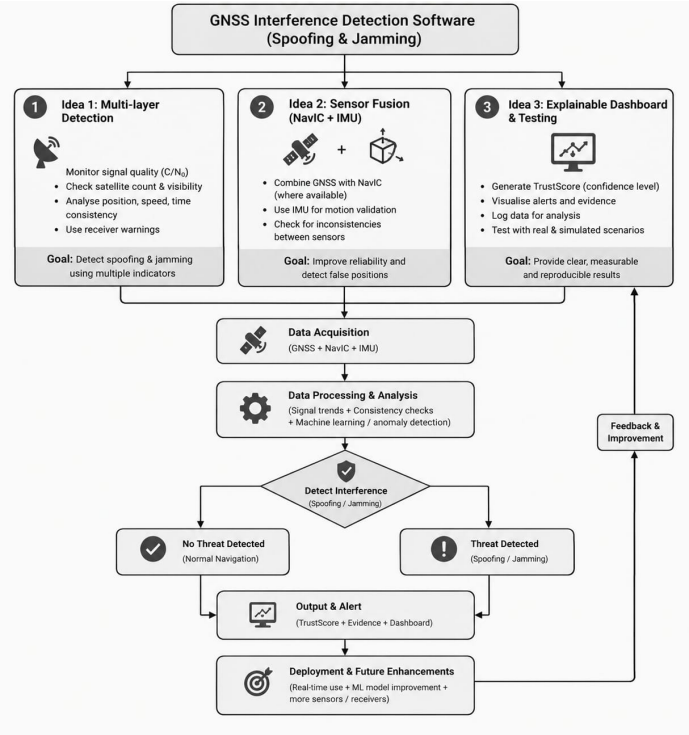


Figure: 1

Tech Stack

Frontend: React.js, HTML, CSS, JavaScript

Backend: Python, FastAPI

Database: SQLite or PostgreSQL

AI/ML, APIs, or Hardware: PyTorch Geometric (GNN), GNSS/RINEX datasets, IMU sensor data

Innovation & Uniqueness

- 1.Our team proposes a software-based GNSS spoofing detector using Graph Neural Networks (GNNs) to identify suspicious satellite signals, rather than relying only on heavy, expensive high-end defense hardware. This makes the approach more practical for existing receivers and smaller drones, where affordable deployment matters.

- **2.GNSS spoofing is an asymmetric threat**: it is easier to spoof a signal than to defend against it, and legacy receivers are difficult and costly to upgrade. A GNN-based model is well suited because the team aims to learn unusual patterns in satellite signals, improve accuracy, reduce false alarms, and enable faster detection and real-time alerts.

- 3.The key differentiator is indigenous, tamper-resistant navigation protection that is practical for current systems: it is designed to strengthen India's defense, aviation, and critical infrastructure by detecting spoofing without depending only on restricted services, GPS augmentation, or expensive hardware-heavy solutions.

Feasibility & Scalability

- Our software works well. Can grow easily across different platforms that rely on navigation. The biggest concerns are the compute costs, trouble connecting with older systems and making sure the validation results are accurate.
- **Feasibility:** the project is explicitly positioned as a software-based model rather than heavy hardware, making it more practical for deployment on existing systems.
- **Scalability:** Our software can scale by moving from a prototype anti-spoofing model to a broader, software-first protection layer across multiple GNSS-dependent sectors and platforms. Its main growth path is from real-time detection on existing receivers to wider deployment in defense, aviation, drones, telecom, and critical infrastructure
- **Risks:** 1. High computational cost-High-end spoofing defense can be expensive and power-hungry 2. Legacy system integration-Many receivers use legacy systems, and upgrades are tedious and economically impractical

Use Cases (compulsorily 3 use cases must be there)

- **Civil aviation safety**:

  user :Airport / airline navigation team

  A flight approaching a busy airport receives a spoofed GNSS signal, and our software detects suspicious navigation data in real time to help prevent wrong positioning.

- **Military maritime navigation**:

  user :Warship / naval defense operator

  A warship operating in a sensitive border area is exposed to fake satellite signals, and the system flags the attack so the vessel can rely on trusted navigation.

- **Drone protection**:

  user :Drone operator / defense drone unit

  A small drone used for surveillance or delivery is targeted by spoofing, and our software helps reject suspicious signals without needing expensive hardware upgrades.

Extra cases

- **Critical infrastructure security**:

  Power grid / essential services control team

  A critical facility depending on GNSS timing or positioning faces fake-signal interference, and the system provides tamper-proof navigation security to protect operations.

Impacts and Benefits

- 1.Airport, airline, naval, drone, telecom, and critical infrastructure teams get a way to detect suspicious GNSS data in real time and act before navigation is corrupted.
- 2. Existing receivers and small drones can gain protection without full hardware replacement, making adoption more practical.
- **3.Social / safety impact:** better protection for aviation, maritime, drone, and essential services operations by reducing the risk of spoofing-related navigation errors.
- 4.**Economic impact:** a software-based approach is more affordable and scalable than expensive high-end defense hardware, which lowers deployment barriers for multiple sectors
- 5. Real-time alerts for suspicious navigation data
- 6.Improved detection accuracy with reduced false alarms through GNN-based pattern learning

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Build resilient navigation infrastructure through GNSS anti-spoofing
SDG 16	Target 16.a	Peace, Justice and Strong Institutions	Strengthen security and institutional resilience against spoofing-based threats

Individual Member Contribution

Member	Contribution or responsibility
Jishnu S	Team leader
Sai Abinav S	powerpoint , uploading documents and reviewing the research paper
Athul Krishna	Refining the ideas and researching publications
Bharath K B	reviewing the research paper and finding gap in the publications which increases accuracy in our software

Yukti ID

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